WAIMLAp 2026

Workshop on Artificial Intelligence and Machine Learning Applications

PUCP

How Do Web-Pretrained DINOv3 Features Extend to Martian Terrain? A Study on AI4Mars

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MOTIVATION & RESEARCH PROBLEM

- Foundation vision models are pretrained on web images, yet they are used on scenes that look nothing like the web, such as the surface of Mars.
- Rovers must tell soil, bedrock, sand and big rock apart; expert labels are scarce and ambiguous.
- **Question:** how do the features of a DINOv3 pretrained on web images extend to the Martian domain (AI4Mars)?

RESEARCH QUESTIONS

- **Layers:** how far does a frozen DINOv3 go, layer by layer, and what do a decoder and fine-tuning add?
- **P1, data:** is the gap one of image appearance?
- **P2, model:** where is it worth adapting, with the same LoRA budget on early, middle or late blocks?
- **P3, errors:** where does it fail and does it know? Expert agreement and calibration.
- Closest work uses frozen DINOv2 on other Mars data and tasks [5]; we found no controlled study of adaptation depth or calibration on AI4Mars.

METHODOLOGY

- **Data:** AI4Mars, Curiosity NavCam, 512 px, 4 classes. 15,564 training images, 500 validation (temporal hold-out), 322 expert test images in three sets M1 / M2 / M3 (labels agreed by at least 1 / 2 / 3 experts).
- **Backbone:** DINOv3 ViT-B/16, frozen (85.7 M parameters, 215 GFLOPs per image, 12 identical blocks). Probe: BatchNorm + 1×1 convolution on one layer's tokens (4.6 k parameters).
- **Heads:** decoder over blocks 3, 6, 9, 12 (3.5 M); last 2 or 4 blocks fine-tuned (14 M / 28 M); LoRA rank 8 on two blocks, early (1–2), middle (6–7) or late (11–12), 221 k parameters.
- **Appearance (P1):** histogram equalisation, CLAHE, per-image standardisation, Gaussian blur.
- **Errors (P3):** mIoU where three experts agree vs where one labelled; calibration (ECE).
- **Statistics:** 3 seeds (2 for LoRA, 1 for the DeepLabV3 baseline); paired bootstrap 95 % CI over blocks of 10 consecutive images; a difference is real if the CI excludes 0 and $|\Delta| \geq 0.01$, equivalent if inside ±0.01. Hypotheses and rules written before the runs.

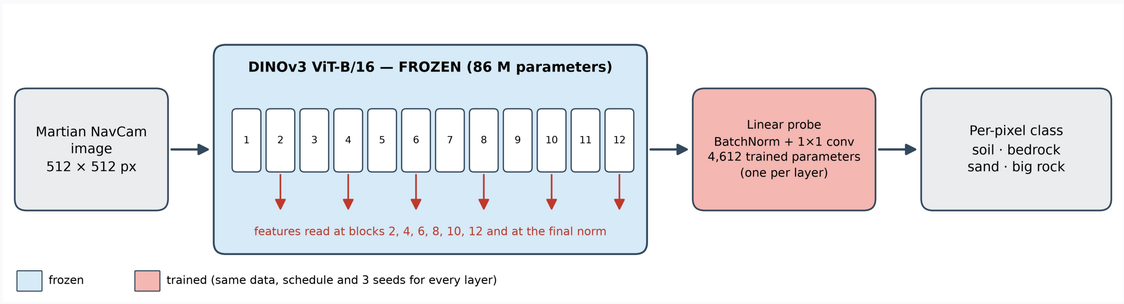


Fig. 1. Frozen DINOv3 ViT-B/16 with one linear probe per tapped layer.

RESULTS 1 · TRANSFER AND LAYERS

- **Transfer:** a linear probe on the final layer reaches mIoU 0.677 / 0.725 on test M1 / M2 (0.125 for always predicting the majority class).
- **Heavier heads add little:** decoder 0.684 / 0.735 and last 4 blocks fine-tuned 0.687 / 0.737, at most +0.012 over the probe.
- **No intermediate layer beats the last;** block 6 already gives 98 % of the final M1 value.
- **Pretraining matters:** a random-init ViT gets 0.264 on M1; DINOv3 is +0.013 [+0.002, +0.025] over DINOv2 on M1, and matching the token grid leaves the gap unchanged.

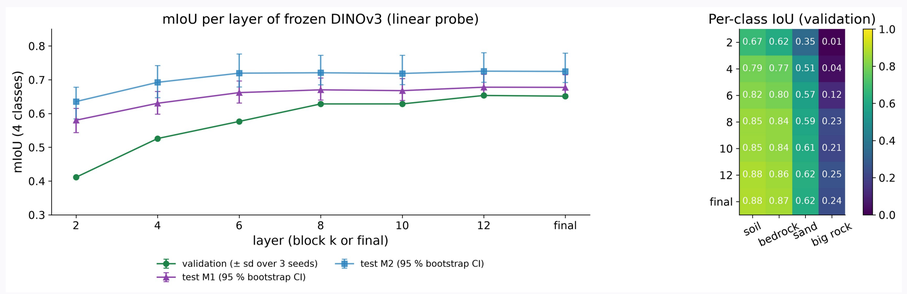


Fig. 2. mIoU per layer of frozen DINOv3 (validation ± sd over 3 seeds; test sets with 95 % bootstrap CI) and per-class IoU.

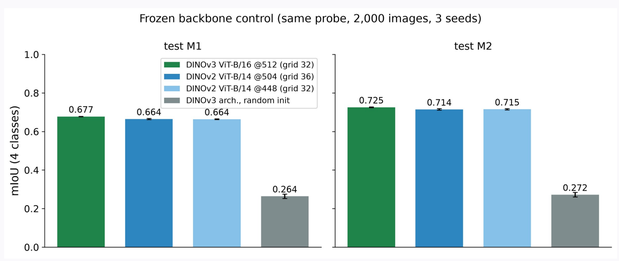


Fig. 3. Frozen backbones, same probe.

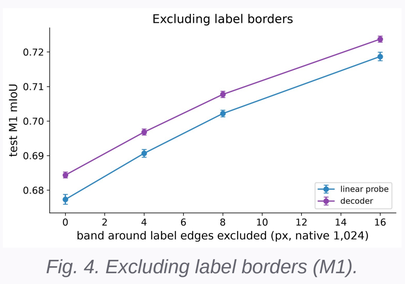


Fig. 4. Excluding label borders (M1).

CONCLUSIONS

- **Web-pretrained DINOv3 features already work on Mars with almost no adaptation:** a 4.6 k-parameter probe is about 0.02 mIoU below the best published end-to-end models and matches a DeepLabV3 trained here, within the statistical margin.
- Neither the appearance of the images nor the depth at which the backbone is adapted moves the score by more than about 0.01 mIoU.
- What limits it is label ambiguity and the rare big-rock class, where the model is confidently wrong. Better labels and class-aware losses come before a heavier model.
- **Limits:** one backbone, one rover, 322 test images (appearance and adaptation depth cannot be decided); LoRA 2 seeds and 2 epochs; DeepLabV3 baseline 1 seed and 3 epochs; M3 holds big rock in only 5 images.
- **Ethics:** public NASA data, noisy crowd-sourced labels; not to be used alone for hazard detection.

RESULTS 2 · APPEARANCE AND ADAPTATION

- **Reference:** the probe reaches 0.674 / 0.723 on M1 / M2, level with DeepLabV3-R50 trained end to end (0.667 / 0.713; M1 difference −0.007, 95 % CI [−0.025, +0.008]), with about 8,600 times fewer trainable parameters. Without pretraining the baseline is clearly behind (0.628).
- **P1, appearance:** CLAHE and standardisation are equivalent to the original images on M1 (+0.001); equalisation is slightly lower (−0.006); blur does not hurt (−0.002). The gap does not look photometric.
- **P2, where to adapt:** LoRA adds +0.005 to +0.007 in every zone, and the zones do not differ (late − early −0.000, [−0.006, +0.015]).

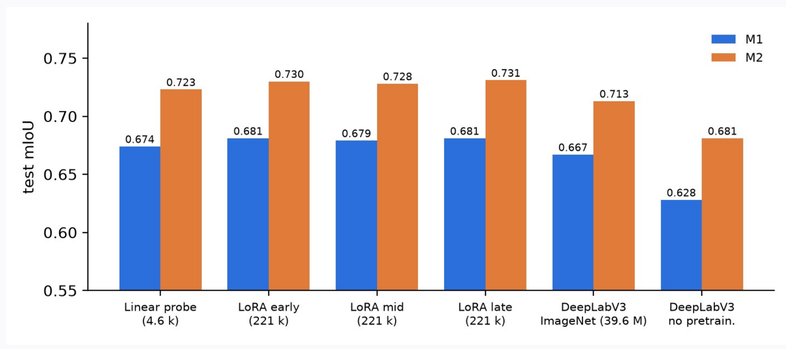


Fig. 5. Test mIoU: probe, LoRA by zone and DeepLabV3 baselines (LoRA 2 seeds; baselines 1 seed, 3 epochs).

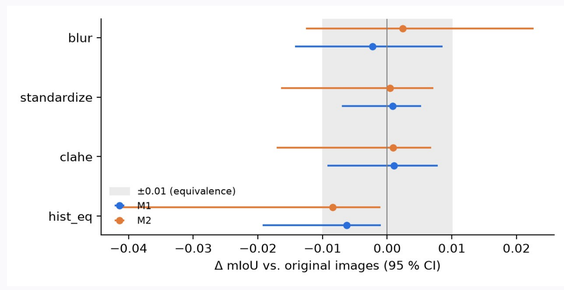


Fig. 6. P1: change in probe mIoU under each appearance transform (grey band: ±0.01).

RESULTS 3 · WHERE IT FAILS

- **Labels:** where three experts agree mIoU is 0.862, where only one labelled it is 0.546 (+0.315 [+0.180, +0.413]). 250 labelled images give 99 % of the 4,000-image probe mIoU, and excluding label borders raises every head by about 0.025.
- **Calibration:** ECE 0.029 overall, but on pixels predicted big rock confidence exceeds accuracy by 0.674 [+0.435, +0.762]; 45 % of big-rock pixels are missed. M3 holds big rock in only 5 images, so it cannot rank models on that class.

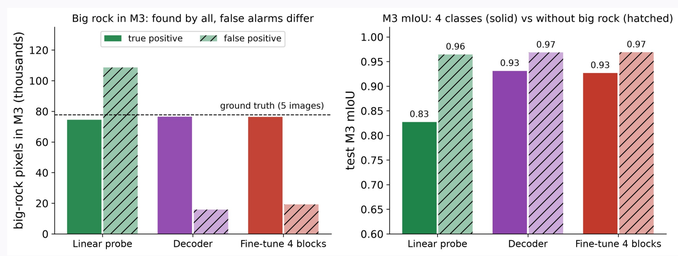


Fig. 7. Big rock in test M3: true and false positives (left); mIoU with and without that class (right).

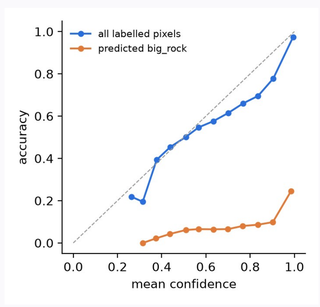


Fig. 8. Calibration of the probe.

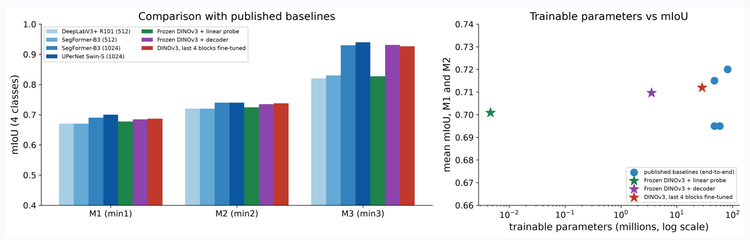


Fig. 9. Comparison with published baselines (M1–M3) and trainable parameters.

LABEL EFFICIENCY AND CONFUSIONS

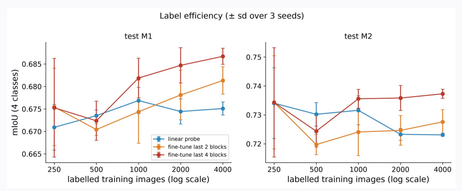


Fig. 10. Label efficiency: probe vs fine-tuning.

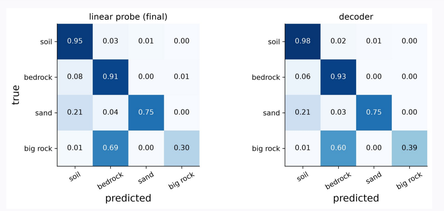


Fig. 11. Row-normalised confusion matrices: probe (left) and decoder (right).

KEY REFERENCES

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[3] F. Mohammad, Y. Li, Y. Gao, “Transformers vs CNNs: Enhancing semantic segmentation for Martian terrain classification,” iSpaRo, 2025.

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[5] P. Durai, G. Doran, “MANTLE: a framework for adaptive in-situ planetary perception using a modular uplink principle,” arXiv:2608.28724, 2026.

[6] E. J. Hu et al., “LoRA: Low-rank adaptation of large language models,” ICLR, 2022.

CONTINUE THE CONVERSATION

Code, notebook and results: https://github.com/gonzaquis29/DinoV3-AI4Mars

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Data and model: AI4Mars (NASA/JPL), DINOv3 (Meta AI). Compute: free-tier Kaggle GPU.

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